Shay Mordechai

Security Researcher | Vulnerability Discovery & Protocol Analysis

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Summary

> Security Researcher specializing in **Low-Level Vulnerability Discovery**, **OS Internals**, and **Reverse Engineering**. Discovered 3 Zero-Days (CVEs pending) via custom AI fuzzing. Deep expertise in memory isolation (Sandbox bypasses, NX evasion) and architectural logic flaws. Passionate about bridging low-level system execution with automated exploit development and vulnerability mitigation.

Professional Experience

DevSecOps Intern | Israel Tax Authority | Aug 2024 – Jul 2025

- Orchestrated transition to Checkmarx One, integrating automated SAST/SCA security gates into 15+ CI/CD pipelines.
- **Vulnerability Triage & Detection:** Developed Python scripts to filter noise and prioritize critical threats (SQLi, DOM XSS, Supply Chain). Guided R&D remediation by translating scan results into defensive coding patterns.

Combat Soldier | Combat Intelligence Collection Corps, IDF (Hesder) | Aug 2016 – Apr 2018 - Executed covert reconnaissance and visual anomaly detection to build actionable operational target profiles.

Education

B.Sc. Computer Science | Jerusalem College of Technology | 2021-2025 | GPA: 89

Coursework: Network Analysis (97), Reverse Engineering (93), Information Security (90).

Continuous Learning: Advanced Malware Analysis (In Progress) – Mastering static/dynamic analysis, IoCs extraction, unpacking, and custom Yara rules for threat detection.

Technical Projects & Research *(Details in Portfolio)*

- **Vulnerability Research (SBCL Compiler)** – Managed full-cycle research via Python fuzzing, uncovering 3 Zero-Days (CVEs pending). Authored documentation for MITRE and National CERT (#11265).
- **OS & Browser Internals Research** – Published deep-dive architectural analysis on V8 JIT vulnerabilities, memory isolation bypasses (RWX allocation, NX flag evasion), and React2Shell execution flows. Evaluated OS-level mitigations including immutable filesystems (Fedora Kinoite) and application sandboxing.
- **Network Protocol Security (DNS)** – Discovered DNS Rebinding vectors targeting local management interfaces; proposed deterministic TTL-hardening mitigations (validated by TP-Link & Palo Alto PSIRT).
- **TLS 1.2 Handshake Simulator** – Engineered a 4,000-line Python tool simulating handshakes and dissecting TCP/IP flows. Used Wireshark/tcpdump for network forensics and testing evasion techniques against Firewalls and IPS/IDS.
- **LeadFlow AI (Zero-Trust SaaS)** – Architected a secure B2B platform on AWS. Eliminated public attack surface via Cloudflare Tunnels and rootless Podman containers.

Technical Skills

- **Security & Research:** OS Internals, Low-Level Vulnerability Discovery, Reverse Engineering, Exploit Development, Fuzzing, Sandbox/Isolation Bypasses, Malware Analysis.
- **Dev & Low-Level: C/C++, x86 Assembly**, Python (Tooling & Automation), Bash, Linux Kernel Concepts.
- **Networking & Cloud:** TCP/IP, DNS, SSL/TLS, AWS, Docker/Podman, Zero-Trust Architecture.